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STUDIES IN HUMANS

Age-related decline of motor unit firing rate in community-dwelling healthy older adults and daily ingestion of fish protein of Alaska pollack: a randomised trial

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ABSTRACT

Neuromuscular properties decline with ageing and low-level physical activities. Alaska pollack protein (APP) is reportedly effective for improving skeletal muscular functions, even if the amount is small and exercise is not conducted. However, it is unclear whether APP intake without an exercise program affects neuromuscular dysfunction in community-dwelling adults conducting normal, everyday activities. This study aimed to investigate changes in motor unit firing properties by adding APP to daily meals in community-dwelling older adults. They were divided into an APP group or a placebo control (CON) group for randomised, double-blind treatment. Participants in APP ($n=15$) and CON ($n=17$) groups ingested 4.5 g/d APP and whey protein, respectively, added to daily meals for 3 months without a specific exercise intervention. High-density surface electromyography of the vastus lateralis to assess the individual motor unit firing rate during 70% of maximum voluntary ramp-up contraction (MVC), maximum knee extensor strength, muscle thickness, echo intensity, and a physical function test were evaluated at the baseline, as well as 1.5 and 3 months after the intervention. While muscle strength, muscle thickness, and motor function were not changed, the motor unit firing rate at 50%–70% of MVC showed a significant decline at 1.5 and 3 months in the CON group, but this decline was not observed in the APP group. These findings suggest that neuromuscular properties show a detectable decline on performing daily activities without a specific exercise intervention, but such a decline was not observed in the APP intake group.

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Introduction

Neuromuscular systems become dysfunctional with ageing, leading to disabilities of mobility and motor functions. Low skeletal muscle mass such as sarcopenia is recognised as a major factor inducing age-relate declines of muscle strength. However, the decline of muscle strength is not always explained solely by the decrease of muscle mass, suggesting that the age-related decline of neural activation is greater in older adults (Manini and Clark 2012; Manini et al. 2013). These reports have important implications for the reduction of age-related decline focusing on the neural system (Aagaard et al. 2010).

An older individual's muscle undergoes morphological and electrophysiological alterations. For example, regarding contractile properties, the tetanic characteristics evaluated by electrical stimulation show age-related changes (Allman and Rice 2004). Also, the afterhyperpolarization duration of the motor unit potential is prolonged in older adults (Piotrkiewicz et al. 2007). These peripheral alterations can cause age-related changes in the central nervous system. Since the duration of afterhyperpolarization is negatively correlated with the motor unit firing rate (Christie and Kamen 2010), it has also been reported that the motor unit firing rate in older adults is lower than that in young adults (Klass et al. 2008; Kallio et al. 2012; Kirk et al. 2021), and declines between middle- and older ages (Piasecki et al. 2021). On considering previous studies and another study (Seki et al. 2007) which reported that the maximal motor

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unit firing rate is reduced after limited physical activity due to immobilisation of muscle, the motor unit firing rate is generally reduced by ageing and limited physical activities. Therefore, it is important to minimise neuromuscular alterations resulting from ageing or low-level physical activities.

To improve or maintain muscle functions, it is important to take appropriate nutrition in conjunction with exercise. Dietary protein intake (1.6 g kg^{-1} day^{-1}) can induce effective improvement of muscle strength and increase skeletal muscle mass (Mitchell et al. 2017). However, in the case of intervention for older adults, it may be appropriate to include a small amount of protein, which can be easily added to daily meals. Even when adding only a small amount (4.5 g day^{-1}) to daily meals, Alaska pollack protein (APP) can induce muscle hypertrophy in humans (Watanabe et al. 2020; Watanabe et al. 2021; Mori et al. 2022).

APP is an effective protein to increase skeletal muscle mass (Mizushige et al. 2010; Kawabata et al. 2015; Morisasa et al. 2022). APP intake without a specific exercise program induced muscle hypertrophy in animal studies (Mizushige et al. 2010; Morisasa et al. 2019). In animal studies, the effect on voluntary neural activation cannot be assessed. On the other hand, in humans, neuromuscular properties are modulated by the combination of APP intake and resistance training (Watanabe et al. 2018; Watanabe et al. 2020; Watanabe et al. 2021). APP intake can increase muscle hypertrophy, and incidentally modulate neural adaptation. A recent study investigating the effects of APP or whey protein intake (4.5 g day^{-1}) without a specific exercise program in older adults found that muscle strength and skeletal muscle mass were increased after APP intake, but not whey protein intake (Mori et al. 2022). However, the effects on the voluntary neural system, which are primary contributors to muscle strength improvement, are unclear.

The purpose of the present study was to investigate changes in neuromuscular functions induced by adding APP to daily meals in community-dwelling older adults in a randomised trial. We hypothesised that neuromuscular functions in older individuals' muscle decline in normal daily lives, and that APP intake could reduce neuromuscular dysfunctions in community-dwelling older adults during 3-month intervention.

Methods

Participants

Thirty-six community-dwelling healthy older adults participated in the present study (10 men and 26 women, age: 73.2 ± 4.4 yrs, height: 157.2 ± 7.0 cm, body mass: 55.1 ± 8.2 kg). They were assigned to two groups for randomised, double-blind, placebo-controlled treatment: placebo control (CON) group (73.7 ± 5.0 yrs, 158.0 ± 7.0 cm, 54.6 ± 7.6 kg) or APP group (72.78 ± 3.6 yrs, 156.4 ± 8.3 cm, 55.6 ± 8.7 kg). Each group consisted of 5 men and 13 women. The experimental procedures and purpose were explained to the participants before they provided informed written consent to participate in the present study. This study was approved by the Research Ethics Committee of Chukyo University (2021-059) and conducted in accordance with the Declaration of Helsinki. The trial was registered at the University Hospital Medical Information Network (UMIN000049568).

Protein supplementation

The participants in both groups ingested protein powder in a cup of soup every day. They were instructed to ingest it at breakfast if possible. Participants in the APP group ingested APP protein powder composed of 5.1 g of protein (4.5 g of APP and 0.6 g of other protein sources), 1.5 g of fat, and 4.0 g of carbohydrate. Those in the CON group ingested whey powder composed of 5.0 g of protein (4.5 g of whey protein and 0.5 g of other protein sources), 1.5 g of fat, and 4.1 g of carbohydrate. The total energy content of both soups was 50 kcal. Nutritional supplementations with both powders were applied as a randomised, double-blind, placebo-controlled treatment. To blind the study, the powders were packed in identical bags and flavourings were controlled by a professional food company. After the intervention, participants reported the number of remaining unused powder bags to calculate the rate of compliance with taking the supplements.

Procedure

The participants visited the laboratory three times before, during and after the intervention (Figure 1). On the first visit, baseline data on the body composition, neuromuscular functions, motor function test, habitual physical activity, and habitual diet were obtained. After 1.5 months, muscle strength, quantity of skeletal muscle mass, and neuromuscular functions were measured again (1.5 mo). After 3 months, the same measurements as conducted at the baseline were performed and participants were interviewed to verify compliance with ingestion of experimental protein (3 mo). The participants were instructed to keep their habitual diets as constant as possible, and not to begin a specific exercise program that increased daily activity.

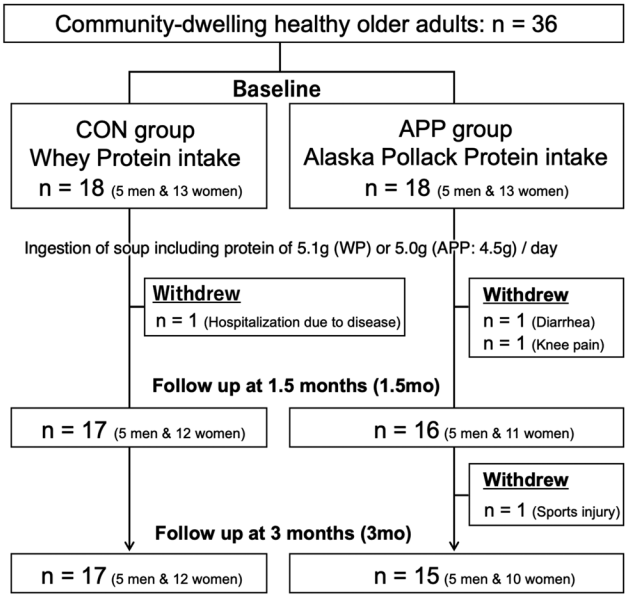

Figure 1. Experimental flow chart. *n*: number of participants.

Body composition

Body mass, skeletal muscle mass, and body fat of the whole body were measured and estimated from bio-impedance measurements (Inbody 430, Inbody Japan Inc., Tokyo, Japan).

Neuromuscular functions

Muscle strength

Participants were seated in a chair with a dynamometer (Takei Scientific Instruments Co., Ltd., Niigata, Japan) fixed to a force transducer (LU-100KSE; Kyowa Electronic Instruments, Aichi, Japan). The hip and knee were flexed at 90 degrees. Before performing maximum voluntary contraction (MVC), they were asked to perform submaximal contractions repeatedly as a warm-up. They performed maximum voluntary isometric knee extension twice. The greater value was selected as the MVC force for further analysis. Maximum knee extension torque was calculated by multiplying the MVC force and arm length, which was determined as the distance between the knee joint axis and force transducer.

Motor unit firing rate

Submaximal ramp-up contraction, consisting of a 14-s increasing phase to 70% of the MVC force level and a 5-s sustained phase at 70% of MVC, was performed. To detect motor units recruited at lower force levels during 70% of MVC ramp-up contraction, a 30% of MVC ramp-up contraction task (15-s increasing phase and 15-s sustained phase) was also performed and used for decomposition analysis (Francic and Holobar 2021). The sustained phase was used only to ensure reliable motor unit identification during analysis. The target force levels among the three visits (baseline, 1.5 mo, and 3 mo) were the same levels based on the MVC value at the baseline. During submaximal ramp-up contraction, high-density surface electromyography (HDsEMG) signals were obtained from the vastus lateralis using an EMG acquisition device (Sessantaquatro, OT Bioelettronica, Torino, Italy) that was synchronised with the force signal. A 64-channel electrode grid (13 × 5 with one missing electrode in the corner; a 1-mm diameter and 8-mm inter-electrode distance; GR08MM1305, OT Bioelettronica) was attached to the skin with a bi-adhesive sheet (KITAD064, OT Bioelettronica) and conductive paste (Elefix Z-181BE, Nihonkohden, Tokyo, Japan). The grid electrodes were attached along the line between the head of the greater trochanter and the superior lateral edge of the patella, and the centre of the grid was placed at the midpoint of the line after the skin was cleaned with alcohol or water. A reference electrode was attached on the tibial tuberosity. Monopolar EMG signals were filtered with a bandpass filter from 10 to 500 Hz and amplified by a factor of 150, sampled at 2000 Hz. Recorded monopolar EMG signals were transferred to analysis software (MATLAB R2019a, MathWorks GK, MA, USA), and individual motor units were identified by the Convolution Kernel Compensation (CKC) technique using Decomposition of Motor Unit Surface EMG (DEMUSE) software (ver. 6.1; The University of Maribor, Slovenia) (Holobar et al. 2009). A single experienced investigator (TH) manually inspected all decomposition results. Any physiologically irregular motor unit discharge (<4 or >50 Hz) and individual motor units with firing rates showing a coefficient of variation over 30% were discarded (Adam and De Luca 2005; Welsh et al. 2007; Hirono et al. 2022). Detected motor units were divided into three groups by the recruitment threshold at each time-point. Recruitment thresholds of motor units were determined from the force level when the motor unit initially discharged. Motor units recruited at less than 20% of MVC were defined as $MU_{0–20}$, those recruited at more than 20% and less than 40% of MVC were defined as $MU_{20–40}$, and those recruited at more than 40% and less than 60% of MVC were defined as $MU_{40–60}$. Firing rates of individual motor units were calculated from the inter-spike intervals. For $MU_{0–20}$, median firing rates during 10%–30%, 30%–50%, and 50%–70% of MVC in the

increasing phase were calculated. For MU_{20-40}, median firing rates during 30%–50% and 50%–70% of MVC were calculated. For MU_{40-60}, the median firing rate during 50%–70% of MVC was calculated.

Ultrasound images

Participants sat relaxed on a chair with their knees flexed at 90°, without their feet contacting the ground. B-mode ultrasound images were obtained from the vastus lateralis muscle using an ultrasound device (MicrUS EXT-1H, TELEMED, Vilnius, Lithuania) with a multifrequency linear array probe (L12-5L40S-3). The equipment settings were as follows: a frequency of 7.5 MHz, dynamic range of 66 dB, gain of 72%, and depth of 70 mm. The probe was positioned at the midpoint between the head of the greater trochanter and the superior lateral edge of the patella with a coupling gel. Longitudinal and transverse images were acquired with the application of minimal pressure. Images were obtained when the bone echo was brightest. Muscle thickness of the vastus lateralis and vastus intermedius was measured as the distance between the superficial fascia of the vastus lateralis and bone edge in longitudinal images. Muscle echo intensity of the vastus lateralis was evaluated in transverse images after they were loaded into software (Image J ver. 1.53k; National Institutes of Health, Bethesda, MD, USA). Echo intensity was computed by the average greyscale value in the region of interest which was determined as wide as possible, excluding the surrounding fascia. Greyscale values ranged from 0 to 255 (black to white), with a higher value indicating a greater amount of fat and fibrous tissue within the muscle (Pillen et al. 2009). Muscle thickness and echo intensity were evaluated as the mean value between two longitudinal and transverse images, respectively.

Chair stand test

Participants repeatedly performed sitting down and standing up from a chair as the 30-s sit-to-stand test. They started from a seated position on a chair whose height was 40 cm, with arms folded across the chest. They were instructed to stand fully upright and return to the seated position as quickly as possible for 30 s. The number of times they fully sat down in 30 s was the score.

Habitual physical activity

Participants wore a triaxial accelerometer (HJA-750C Active style Pro; Omron Healthcare, Kyoto, Japan) for 1 week at both the beginning and end of the intervention period. If one day's data included an extreme outlier, the data were discarded. Total energy consumption and total step counts were calculated per day.

Habitual diet

A dietary survey was evaluated by nutritionists (SU and YM) with a licence from the Ministry of Health, Labour and Welfare in Japan. Total energy, carbohydrate, protein, and fat in daily diets including the experimental supplements were evaluated by a brief-type self-administered diet history questionnaire (BDHQ) (Kobayashi et al. 2012). These nutrition values were normalised by the individual body mass to estimate the individual nutritional conditions. This questionnaire was applied at baseline measurement.

Statistical analysis

All statistical analyses were performed with SPSS (Statistical Package for the Social Sciences ver. 25.0, IBM Japan Inc., Tokyo, Japan). The normality of data distributions for all variables were confirmed by the Shapiro-Wilk test. When data were normally distributed, repeated measures two-way ANOVAs with interactions within- and between-factors were used to compare time-points between groups for variables of body composition, muscle strength, muscle thickness, echo intensity, and physical activity. When data were non-normally distributed, Friedman tests or Wilcoxon signed-rank tests in each group were used to compare time-points for variables with the same outcomes. Nutritional parameters were compared between groups using *t*-tests or Mann–Whitney *U* tests based on the results of normality tests. Because the data of the motor unit firing rate were almost non-normally distributed, Kruskal–Wallis tests were used for each ingestion group, each motor unit group, and each analysis interval of force level to compare time-points. If interactions or main effects were found in each statistical analysis, post-hoc tests were performed with Bonferroni corrections. The significance level was set at 0.050.

Results

Figure 1 shows a flow chart of the present experiment. The total numbers of participants who completed the 3-month intervention were 17 in the CON group (5 men and 12 women, age; 73.7 ± 5.2 yrs, height; 158.3 ± 7.1 cm) and 15 in the APP group (5 men and 10 women, age; 72.7 ± 3.7 yrs, height; 156.5 ± 8.4). Body mass, skeletal muscle mass, and body fat mass

are presented in Table 1. The two-way ANOVA for skeletal muscle mass revealed a main effect of time, but post-hoc tests did not identify any significance. Body mass and body fat mass did not significantly change. Nutritional parameters did not differ between the two groups (Table 2).

Knee extension, thigh muscle thickness, echo intensity, 30-s sit-to-stand test, and physical activity outcomes are described in Table 1. Two-way ANOVAs for muscle strength, muscle thickness, and total energy consumption did not reveal interaction or main effects. The Friedman test revealed a significant change in echo intensity in the CON group, but not the APP group. Post-hoc tests identified a significant difference between the baseline and 1.5 mo. Wilcoxon tests for 30-s sit-to-stand tests and step counts revealed no significant differences.

The total number of detected motor units was 891. The numbers of $MU_{0\text{-}20}$ at the baseline, 1.5 mo, and 3 mo were 56, 50, and 59 in the CON group, and 39, 34, and 35 in the APP group, respectively. The numbers of $MU_{20\text{–}40}$ at the baseline, 1.5 mo, and 3 mo were 66, 67, and 73 in the CON group, and 46, 61, and 51 in the APP group, respectively. The numbers of $MU_{40\text{–}60}$ at the baseline, 1.5 mo, and 3 mo were 43, 52, and 43 in the CON group, and 33, 37, and 46 in the APP group, respectively. Motor unit firing rates in each ingestion group, each motor unit group, and at each exerted force level are shown in Figure 2. In the CON group, there were no significant differences among time-points in motor unit firing rates of $MU_{0\text{-}20}$ and $MU_{20\text{–}40}$ at 10%–30% and 30%–50% of MVC based on Kruskal Wallis tests ($p = .193$–$.493$). At 50%–70% of MVC, Kruskal Wallis tests revealed significant differences in $MU_{0\text{–}20}$ ($p = .014$, Cohen's $d = 0.82$) and $MU_{40\text{–}60}$ ($p < .001$, Cohen's $d = 0.67$), but not $MU_{20\text{–}40}$ ($p = .544$). The firing rate of $MU_{0\text{-}20}$ at 1.5 mo was significantly lower than that at the baseline ($p = .011$), but there was no difference between the baseline and 3 mo ($p = .804$) or between 1.5 mo and 3 mo ($p = .196$). Firing rates of $MU_{40\text{–}60}$ at 1.5 mo and 3 mo were significantly lower than that at the baseline ($p < .001$ and $p = .007$), but there was no difference between 1.5 and 3 mo ($p = 1.00$). On the other hand, in the APP group, Kruskal Wallis tests revealed no significance difference in each motor unit group or at each force level ($p = .245$–$.982$).

Discussion

The present study investigated the chronic effects of daily APP intake on neuromuscular properties in community-dwelling older adults. The participants did not conduct specific exercise programs; solely performing normal daily activities. Muscle strength and physical functions remained unchanged after the 3-month ingestion in both groups. Focusing on neural activities, we identified a significant decrease in the firing rate of motor units with a high recruitment threshold in the CON group, but did not observe such a significant change in the APP group (Figure 2). The results suggest that the decrease of neural

Table 1. Results of two-way ANOVAs or Friedman tests for the characteristics in CON ($n = 17$) and APP ($n = 15$) groups.

		Baseline	1.5 months	3 months	Interaction	Time effect	Group effect
Body mass [kg]	CON	54.8 ± 8.0	54.9 ± 7.8	54.9 ± 7.7	$F = 0.15$	$F = 0.97$	$F = 0.02$
	APP	55.2 ± 9.7	55.4 ± 10.0	55.3 ± 10.0	$\eta^2 = 0.01$	$\eta^2 = 0.03$	$\eta^2 < 0.01$
					$p = .787$	$p = .360$	$p = .904$
Skeletal muscle mass [kg]	CON	21.8 ± 4.0	22.0 ± 3.8	21.7 ± 3.9	$F = 1.69$	$F = 3.32$	$F = 0.003$
	APP	21.6 ± 5.0	21.8 ± 5.1	21.7 ± 5.2	$\eta^2 = 0.05$	$\eta^2 = 0.10$	$\eta^2 < 0.01$
					$p = .193$	$p = .045$	$p = .954$
Body fat mass [kg]	CON	14.4 ± 3.4	14.3 ± 3.2*	14.8 ± 3.2*	–	$p = .016$ [a]	–
	APP	15.1 ± 6.3	15.1 ± 6.2	15.0 ± 6.2	–	$p = .886$ [a]	–
Knee extension strength [Nm]	CON	107.7 ± 35.9	111.0 ± 32.6	110.3 ± 34.0	$F = 0.33$	$F = 1.98$	$F = 0.41$
	APP	101.5 ± 28.4	104.2 ± 29.9	101.8 ± 28.3	$\eta^2 = 0.01$	$\eta^2 = 0.06$	$\eta^2 = 0.01$
					$p = .720$	$p = .147$	$p = .526$
Thigh muscle thickness [mm]	CON	41.0 ± 7.2	39.9 ± 6.9	41.9 ± 7.0	$F = 1.91$	$F = 1.76$	$F = 0.02$
	APP	41.0 ± 7.8	41.4 ± 7.4	41.4 ± 7.3	$\eta^2 = 0.06$	$\eta^2 = 0.06$	$\eta^2 < 0.001$
					$p = .156$	$p = .181$	$p = .882$
Echo intensity [a.u.]	CON	59.5 ± 7.5*	63.9 ± 7.5*	61.2 ± 7.9	–	$p = .007$ [a]	–
	APP	58.3 ± 9.4	60.1 ± 7.3	59.8 ± 7.1	–	$p = .420$ [a]	–
30-s sit-to-stand test [times]	CON	25 ± 6		24 ± 6	–	$p = .421$ [b]	–
	APP	28 ± 7		28 ± 7	–	$p = .710$ [c]	–
Total energy consumption [kcal/day]	CON	1814 ± 266		1796 ± 252	$F = 0.79$	$F = 3.87$	$F = 0.04$
	APP	1852 ± 350		1802 ± 357	$\eta^2 = 0.03$	$\eta^2 = 0.11$	$\eta^2 < 0.01$
					$p = .382$	$p = .059$	$p = .838$
Step counts [/day]	CON	7178 ± 4513		6195 ± 4353	–	$p = .062$ [c]	–
	APP	7885 ± 3469		7184 ± 3179	–	$p = .256$ [c]	–

Mean ± standard deviation are presented. Split-plot two-way analyses of variance (ANOVAs) were performed. CON: whey protein intake group as control. APP: Alaska pollack protein intake group. The letters "a", "b", and "c" show p-values from the Friedman test, paired t-test, and Wilcoxon signed rank test, respectively. Significant differences calculated by paired tests between those with * ($p < .05$).

activities that occurs during normal daily lives in older adults was not observed in the APP group.

Just as age-related decline of the neural system is often reported (Kamen and Knight 2004; McNeil and Rice 2018; Jones et al. 2022), the motor unit firing rate, which was assessed as the neural system in the present study, significantly decreased over the 3 months (Figure 2). Since the present study did not involve the addition of specific additional exercise to daily activity, the change in the motor unit firing rate could not be caused by exercise adaptation; rather, it would be caused by natural age-related decline. The firing rate of motor units with a relatively high threshold, during high-intensity contraction (50%–70% of MVC), significantly declined during the intervention in the CON group (Figure 2). The declines in the firing rate during high-intensity contraction are considered to be specific changes depending on the contraction intensity. In typical daily activities, people do not perform such high-intensity contractions; therefore, the motor unit firing rate during high-intensity contraction might be specifically altered. The present results show contraction intensity-dependent alteration due to normal/sedentary lives, which, to the best of our knowledge, is a novel finding.

Table 2. Nutritional parameters.

	CON	APP	*p* value
Energy [kcal/day/BW]	37.3 ± 12.0	38.3 ± 11.2	.811[a]
Carbohydrates [g/day/BW]	4.8 ± 1.9	4.6 ± 1.6	.764[a]
Protein [g/day/BW]	1.7 ± 0.6	1.6 ± 0.5	.551[b]
Fat [g/day/BW]	1.1 ± 0.4	1.3 ± 0.5	.526[b]

Mean ± standard deviation are presented. The letter "a" shows the *p*-value from the paired t-test. The letter "b" shows the *p*-value from the Mann–Whitney test. CON: whey protein intake group as control. APP: Alaska pollack protein intake group. BW: body weight.

An age-related decline of the neuromuscular system is particularly evident in type II fibres (Edström and Larsson 1987; Doherty and Brown 1993; Rowan et al. 2011). Denervation and reinnervation are repeated with ageing (Fuertes-Alvarez and Izeta 2021; Jones et al. 2022), resulting in an increase in the size of motor units (Piasecki et al. 2016). According to the fact that motor units with a large size and type II fibres are related to motor units with a high recruitment threshold, the present results show that the firing rate of $MU_{40–60,}$ belonging to a relatively high recruitment threshold group, declined with ageing, even over a 3-month period, due to age-related neuromuscular dysfunction. Although ultrasound images

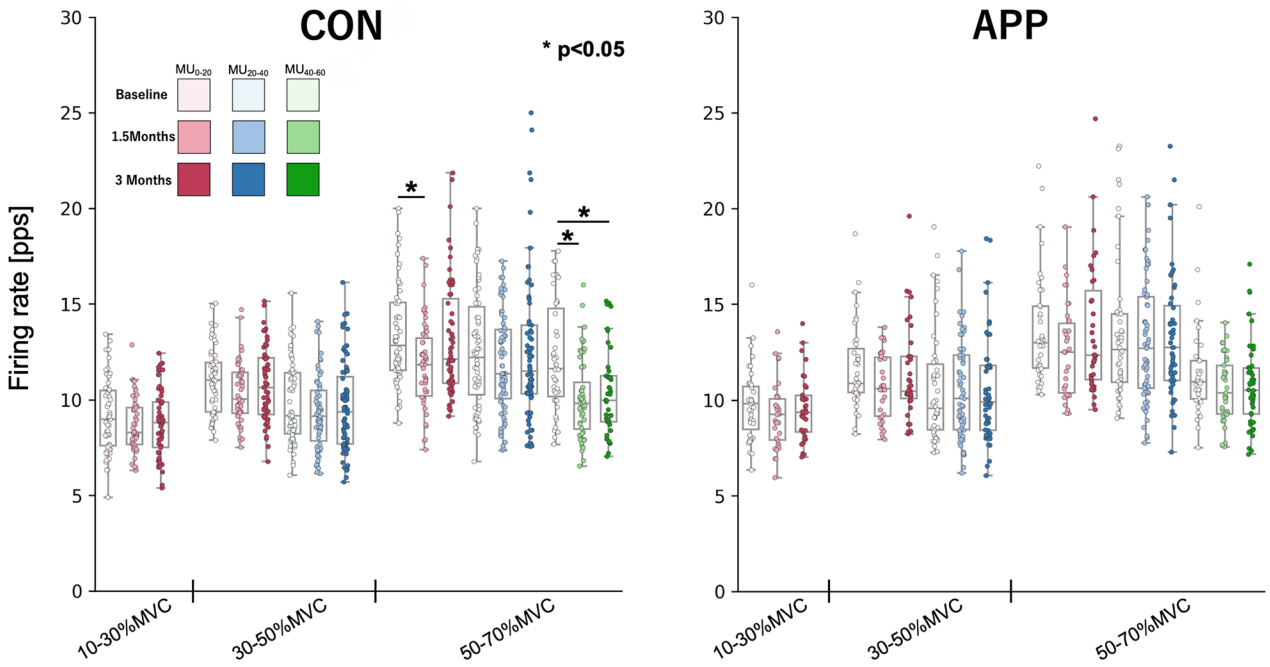

Figure 2. Motor unit (MU) firing rate in each ingestion group, with each motor unit group classified based on the recruitment threshold. Light, medium, and dark red represent motor units recruited at less than 20% of maximum voluntary contraction (MVC) ($MU_{0–20}$) at the baseline, 1.5 months, and 3 months, respectively. Light, medium, and dark blue represent motor units recruited at more than 20% and less than 40% of MVC ($MU_{20–40}$) at the baseline, 1.5 months, and 3 months, respectively. Light, medium, and dark green represent motor units recruited at more than 40% and less than 60% of MVC ($MU_{40–60}$) at the baseline, 1.5 months, and 3 months, respectively. Kruskal–Wallis tests were performed in each motor unit group and at each exerted force level to compare time-points. If the tests found main effects, a post-hoc test involving the Mann–Whitney *U* test with Bonferroni correction was performed. CON: whey protein intake group as control. APP: Alaska pollack protein intake group. The asterisk represents a significant difference based on the post-hoc test ($p < .05$).

and bioimpedance measurements, which gave an overview of muscle, could not detect an increase in muscle mass (Table 1), detectable motor unit firing properties, forming a pathway from the central nervous system, declined. This was an interesting finding, revealing an age-related neural decline in high recruitment threshold motor units and during high-intensity contraction.

The findings show that $MU_{0\text{-}20}$, with a relatively low threshold, significantly fired at a lower frequency at 1.5 mo, but the change did not remain at 3 mo. A change in motor unit firing properties with a low recruitment threshold was observed following resistance training (Watanabe et al. 2018; Hirono et al. 2023) and age-related differences were noted (Watanabe et al. 2016), suggesting that motor units with a low recruitment threshold can exhibit low-level plasticity. Therefore, the present results indicate partial changes in relatively low-threshold motor unit firing.

In the present study, a significant decline in the motor unit firing rate was not observed in the APP group (Figure 2). In a human study, it was recently reported that muscle strength and skeletal muscle mass were increased after APP intake with a small amount (4.5 g d^{-1}), but not whey protein intake (Mori et al. 2022), suggesting that such a small amount of APP could suppress neuromuscular degeneration in the present study because APP intake can promote the concentrations of muscle protein synthesis factors (Morisasa et al. 2022). Previous studies reported that APP intake can induce muscle hypertrophy in animal studies (Mizushige et al. 2010; Kawabata et al. 2015; Morisasa et al. 2019), and those that did not expose the animals to exercise intervention reported the expression of type II protein due to APP intake (Morisasa et al. 2022), as well as significant muscle hypertrophy induced by its intake (Morisasa et al. 2019). Based on the present study, it is suggested that APP intake has a low-level impact on muscle morphology that is not detectable using bioimpedance measurement or ultrasound imaging (Table 1). Type II fibre atrophy is marked in older muscle (Edström and Larsson 1987; Doherty and Brown 1993; Rowan et al. 2011), and APP intake can induce type II fibre hypertrophy (Morisasa et al. 2022). Since motor units with a high recruitment threshold are related to type II fibres, APP intake may have prevented the atrophy of type II fibres and maintained neural excitability in motor units with a high recruitment threshold (such as $MU_{40\text{–}60}$) in the present study. There were limitations in that the measurements of ultrasound and bioimpedance cannot detect micro-muscular adaptations; nevertheless, the current results whereby APP intake reduced only detectable neural dysfunctions, could be important findings to ameliorate age-related neuromuscular dysfunctions, because it has been suggested that such dysfunctions could precede muscle atrophy and muscle weakness (Piasecki et al. 2016).

Although a previous study reported significant improvements of muscle strength and skeletal muscle mass without a specific exercise program (Mori et al. 2022), participants in the APP intake group of the previous study ingested protein of 1.09 ± 0.30 g/d/body weight (BW) and those in the control group ingested 1.12 ± 0.35 g/d/BW in their habitual diets. The previous participants ingested additional protein of APP or whey of 4.5 g/d; nevertheless, muscle in the APP group increased after the intervention, but not that in the control group. The individuals in the current study ingested protein of 1.5–1.6 g/d/BW in their typical diet (Table 2), indicating that the current population consumed daily meals rich in protein. The previous results (Mori et al. 2022) differ from the present ones in that muscle strength/muscle mass were not changed during the intervention, because it is possible that habitual amounts of protein intake differ among populations. Even though our population habitually ate a diet rich in protein, the motor unit firing rate declined in the CON group, while APP intake could prevent such a decline. This suggests that APP intake might have an effect even in older adults who already consume a diet rich in protein.

A significant reduction in firing rate was observed in the CON group (Figure 2), but we should consider factors other than natural age-related changes. Focusing on the firing rate at baseline, it appeared to be higher in the CON group compared to the APP group. The firing rates at 1.5 mo and 3 mo in the CON group were similar to those in the APP group, suggesting that the observed reduction in the firing rate in the CON group might be influenced by factors other than natural age-related changes, such as characteristics of the participants in the CON group. We did not verify these factors in the present analysis, so the interpretation of the results might be approached with caution.

Body fat mass and echo intensity changed during the intervention in the CON group (Table 1). This intervention was conducted in winter, when there is an increase in fat (Tan et al. 2022). Hence, variables may change during the intervention. Because APP has an effect of inhibiting visceral body fat accumulation in rats (Mizushige et al. 2010; Morisasa et al. 2019), we consider that the influence of changes in body fat and intramuscular adipose tissue might be minimal in the APP group. It is a limitation that we evaluated

only global muscular fat accumulation using bioimpedance and ultrasound images, and so the mechanism of fat accumulation in human muscle remains unclear.

This study has some limitations. First, the sample size was small. Second, we did not strictly control the participants' diet and exercise habits during the intervention. However, we instructed the participants to maintain their habits as consistently as possible. Additionally, we did not include habit data in the present statistical model. Such detail habits might be useful for understanding the changes in the neuromuscular properties.

In summary, the present study involved the addition of 4.5 g of APP or whey protein to habitual daily diets of community-dwelling older adults without a specific resistance training program for 3 months, for randomised, double-blind, placebo-controlled treatment. As a results, the firing rate of motor units with a relatively high recruitment threshold declined during high-intensity contraction in only the CON group, whereas such a decline was not observed in the APP group. This suggests that an age-related decline in neural components can be observed even within a short period of 3 months in community-dwelling older adults without an additional specific resistance training program, but such an age-related decline was not observed in the APP intake group.

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References

Aagaard P, Suetta C, Caserotti P, Magnusson SP, Kjaer M. 2010. Role of the nervous system in sarcopenia and muscle atrophy with aging: strength training as a countermeasure. Scand J Med Sci Sports. 20(1):49–64. doi: 10.1111/j.1600-0838.2009.01084.x.

Adam A, De Luca CJ. 2005. Firing rates of motor units in human vastus lateralis muscle during fatiguing isometric contractions. J Appl Physiol. 99(1):268–280. doi: 10.1152/japplphysiol.01344.2004.

Allman BL, Rice CL. 2004. An age-related shift in the force-frequency relationship affects quadriceps fatigability in old adults. J Appl Physiol. 96(3):1026–1032. doi: 10.1152/japplphysiol.00991.2003.

Christie A, Kamen G. 2010. Short-term training adaptations in maximal motor unit firing rates and afterhyperpolarization duration. Muscle Nerve. 41(5):651–660. doi: 10.1002/mus.21539.

Doherty TJ, Brown WF. 1993. The estimated numbers and relative sizes of thenar motor units as selected by multiple point stimulation in young and older adults. Muscle Nerve. 16(4):355–366. doi: 10.1002/mus.880160404.

Edström L, Larsson L. 1987. Effects of age on contractile and enzyme-histochemical properties of fast- and slow-twitch single motor units in the rat. J Physiol. 392(1):129–145. doi: 10.1113/jphysiol.1987.sp016773.

Francic A, Holobar A. 2021. On the reuse of motor unit filters in high density surface electromyograms recorded at different contraction levels. IEEE Access. 9:115227–115236. doi: 10.1109/ACCESS.2021.3104762.

Fuertes-Alvarez S, Izeta A. 2021. Terminal Schwann cell aging: implications for age-associated neuromuscular dysfunction. Aging Dis. 12(2):494–514. doi: 10.14336/AD.2020.0708.

Hirono T, Kunugi S, Yoshimura A, Holobar A, Watanabe K. 2022. Acute changes in motor unit discharge property after concentric versus eccentric contraction exercise in knee extensor. J Electromyogr Kinesiol. 67:102704. doi: 10.1016/j.jelekin.2022.102704.

Hirono T, Kunugi S, Yoshimura A, Ueda S, Goto R, Akatsu H, Watanabe K. 2023. Effects of home-based bodyweight squat training on neuromuscular properties in community-dwelling older adults. Aging Clin Exp Res. 35(5):1043–1053. doi: 10.1007/s40520-023-02370-6.

Holobar A, Farina D, Gazzoni M, Merletti R, Zazula D. 2009. Estimating motor unit discharge patterns from high-density surface electromyogram. Clin Neurophysiol. 120(3):551–562. doi: 10.1016/j.clinph.2008.10.160.

Jones EJ, Chiou SY, Atherton PJ, Phillips BE, Piasecki M. 2022. Ageing and exercise-induced motor unit remodelling. J Physiol. 600(8):1839–1849. doi: 10.1113/JP281726.

Kallio J, Søgaard K, Avela J, Komi P, Selänne H, Linnamo V. 2012. Age-related decreases in motor unit discharge rate and force control during isometric plantar flexion. J Electromyogr Kinesiol. 22(6):983–989. doi: 10.1016/j.jelekin.2012.05.009.

Kamen G, Knight CA. 2004. Training-related adaptations in motor unit discharge rate in young and older adults. J Gerontol A Biol Sci Med Sci. 59(12):1334–1338. doi: 10.1093/gerona/59.12.1334.

Kawabata F, Mizushige T, Uozumi K, Hayamizu K, Han L, Tsuji T, Kishida T. 2015. Fish protein intake induces fast-muscle hypertrophy and reduces liver lipids and serum glucose levels in rats. Biosci Biotechnol Biochem. 79(1):109–116. doi: 10.1080/09168451.2014.951025.

Kirk EA, Christie AD, Knight CA, Rice CL. 2021. Motor unit firing rates during constant isometric contraction: establishing and comparing an age-related pattern among muscles. J Appl Physiol. 130(6):1903–1914. doi: 10.1152/japplphysiol.01047.2020.

Klass M, Baudry S, Duchateau J. 2008. Age-related decline in rate of torque development is accompanied by lower maximal motor unit discharge frequency during fast contractions. J Appl Physiol. 104(3):739–746. doi: 10.1152/japplphysiol.00550.2007.

Kobayashi S, Honda S, Murakami K, Sasaki S, Okubo H, Hirota N, Notsu A, Fukui M, Date C. 2012. Both comprehensive and brief self-administered diet history questionnaires satisfactorily rank nutrient intakes in Japanese adults. J Epidemiol. 22(2):151–159. doi: 10.2188/jea.je20110075.

Manini TM, Clark BC. 2012. Dynapenia and aging: an update. J Gerontol A Biol Sci Med Sci. 67(1):28–40. doi: 10.1093/gerona/glr010.

Manini TM, Hong SL, Clark BC. 2013. Aging and muscle: a neuron's perspective. Curr Opin Clin Nutr Metab Care. 16(1):21–26. doi: 10.1097/MCO.0b013e32835b5880.

McNeil CJ, Rice CL. 2018. Neuromuscular adaptations to healthy aging. Appl Physiol Nutr Metab. 43(11):1158–1165. doi: 10.1139/apnm-2018-0327.

Mitchell CJ, Milan AM, Mitchell SM, Zeng N, Ramzan F, Sharma P, Knowles SO, Roy NC, Sjödin A, Wagner K-H, et al. 2017. The effects of dietary protein intake on appendicular lean mass and muscle function in elderly men: a 10-wk randomized controlled trial. Am J Clin Nutr. 106(6):1375–1383. doi: 10.3945/ajcn.117.160325.

Mizushige T, Kawabata F, Uozumi K, Tsuji T, Kishida T, Ebihara K. 2010. Fast-twitch muscle hypertrophy partly induces lipid accumulation inhibition with Alaska pollack protein intake in rats. Biomed Res. 31(6):347–352. doi: 10.2220/biomedres.31.347.

Mori H, Tokuda Y, Yoshida E, Uchida K, Matsuhisa M. 2022. Chronic intake of a meal including Alaska pollack protein increases skeletal muscle mass and strength in healthy older women: a double-blind randomized controlled trial. J Nutr. 152(12):2761–2770.

Morisasa M, Goto-Inoue N, Sato T, Machida K, Fujitani M, Kishida T, Uchida K, Mori T. 2019. Investigation of the lipid changes that occur in hypertrophic muscle due to fish protein-feeding using mass spectrometry imaging. J Oleo Sci. 68(2):141–148. doi: 10.5650/jos.ess18193.

Morisasa M, Yoshida E, Fujitani M, Kimura K, Uchida K, Kishida T, Mori T, Goto-Inoue N. 2022. Fish protein promotes skeletal muscle hypertrophy via the Akt/mTOR signaling pathways. J Nutr Sci Vitaminol. 68(1):23–31. doi: 10.3177/jnsv.68.23.

Piasecki J, Inns TB, Bass JJ, Scott R, Stashuk DW, Phillips BE, Atherton PJ, Piasecki M. 2021. Influence of sex on the age-related adaptations of neuromuscular function and motor unit properties in elite masters athletes. J Physiol. 599(1):193–205. doi: 10.1113/JP280679.

Piasecki M, Ireland A, Jones DA, McPhee JS. 2016. Age-dependent motor unit remodelling in human limb muscles. Biogerontology. 17(3):485–496. doi: 10.1007/s10522-015-9627-3.

Pillen S, Tak RO, Zwarts MJ, Lammens MM, Verrijp KN, Arts IM, van der Laak JA, Hoogerbrugge PM, van Engelen BG, Verrips A. 2009. Skeletal muscle ultrasound: correlation between fibrous tissue and echo intensity. Ultrasound Med Biol. 35(3):443–446. doi: 10.1016/j.ultrasmedbio.2008.09.016.

Piotrkiewicz M, Kudina L, Mierzejewska J, Jakubiec M, Hausmanowa-Petrusewicz I. 2007. Age-related change in duration of afterhyperpolarization of human motoneurones. J Physiol. 585(Pt 2):483–490. doi: 10.1113/jphysiol.2007.142356.

Rowan SL, Purves-Smith FM, Solbak NM, Hepple RT. 2011. Accumulation of severely atrophic myofibers marks the acceleration of sarcopenia in slow and fast twitch muscles. Exp Gerontol. 46(8):660–669. doi: 10.1016/j.exger.2011.03.005.

Seki K, Kizuka T, Yamada H. 2007. Reduction in maximal firing rate of motoneurons after 1-week immobilization of finger muscle in human subjects. J Electromyogr Kinesiol. 17(2):113–120. doi: 10.1016/j.jelekin.2005.10.008.

Tan SY, Curtis AR, Leech RM, Ridgers ND, Crawford D, McNaughton SA. 2022. A systematic review of temporal body weight and dietary intake patterns in adults: implications on future public health nutrition interventions to promote healthy weight. Eur J Nutr. 61(5):2255–2278. doi: 10.1007/s00394-021-02791-x.

Watanabe K, Holobar A, Kouzaki M, Ogawa M, Akima H, Moritani T. 2016. Age-related changes in motor unit firing pattern of vastus lateralis muscle during low-moderate contraction. Age (Dordr). 38(3):48. doi: 10.1007/s11357-016-9915-0.

Watanabe K, Holobar A, Mita Y, Kouzaki M, Ogawa M, Akima H, Moritani T. 2018. Effect of resistance training and fish protein intake on motor unit firing pattern and motor function of elderly. Front Physiol. 9:1733. doi: 10.3389/fphys.2018.01733.

Watanabe K, Holobar A, Mita Y, Tomita A, Yoshiko A, Kouzaki M, Uchida K, Moritani T. 2020. Modulation of neural and muscular adaptation processes during resistance training by fish protein ingestions in older adults. J Gerontol A Biol Sci Med Sci. 75(5):867–874. doi: 10.1093/gerona/glz215.

Watanabe K, Holobar A, Uchida K, Mita Y. 2021. Fish protein ingestion induces neural, but not muscular adaptations, following resistance training in young adults. Front Nutr. 8:645747. doi: 10.3389/fnut.2021.645747.

Welsh SJ, Dinenno DV, Tracy BL. 2007. Variability of quadriceps femoris motor neuron discharge and muscle force in human aging. Exp Brain Res. 179(2):219–233. doi: 10.1007/s00221-006-0785-z.